02-620 Week 9

Machine Learning for Scientists

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Dimensionality Reduction

Data in High Dimensional Space

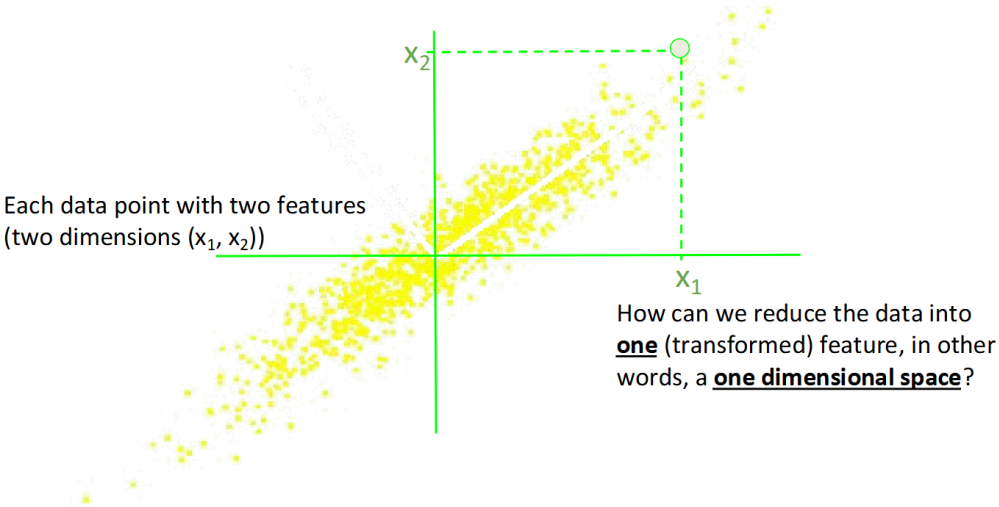
- High-dimensional data
  - Large number of features (large K)
  - Difficult to visualize
  - Difficult to fit a large model on data with many features
- Can we reduce the dimension of the data?
  - For easy visualization
  - For effective model learning

What is Dimensionality Reduction?

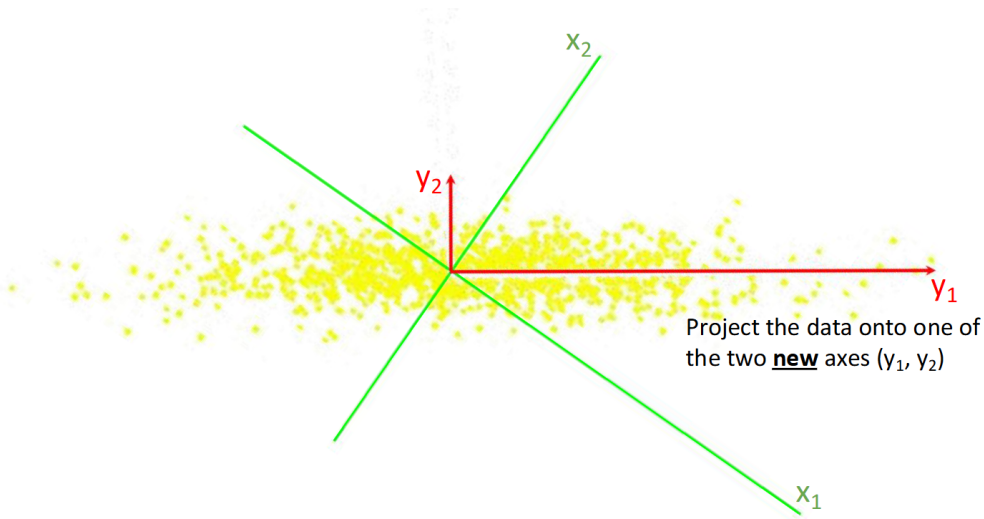
- Dimensionality reduction is a type of unsupervised learning
- We try to reduce the N sample $\times$ K feature training data to a N sample $\times$ M feature training data, where ($M << K$).
- Easy to visualize in 2 or 3 dimensional space
- Transformed features capture most important dimensions
- May capture the low-dimensional structure of the data

Principal Component Analysis (PCA)

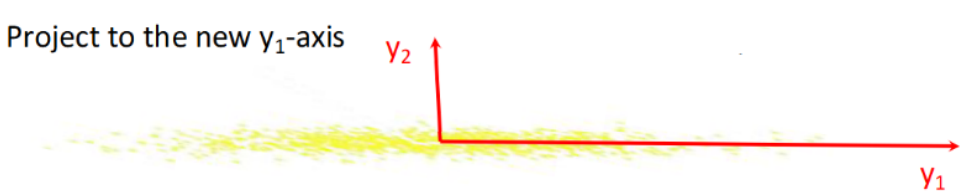
Consider reducing 2-dim data into 1-dim data.



Now, we can create new axes, y_1 and y_2 that also represent the data on the plane using some linear transformation. We rotate this new basis.



Finally, we project to one axes. In this case, projecting onto y_1 is better since the data has **higher variability** along y_1.



The result is that we now have less dimensions, but the data is still represented well (albeit not as well compared to by both features)

How to execute PCA

1. Find the principal components. (linear algebra: eigen-decomposition)
2. Project the data onto the principle components of your choice. (linear algebra: vector inner-product)

Aside: Eigen-decomposition

$$S = UDU^T$$

- S is a $K \times K$ symmetric square matrix
- U is a $K \times K$ square matrix
- D is a $K \times K$ diagonal matrix

In code, we can do this by calling the `numpy.linalg.eigh` function in Python.

- U and D are our eigenvalues and eigenvectors, respectively.
- $U = [u_1, \cdots, u_K] \in \mathbb{R}^{K \times K}$. $u_1, \cdots, u_K \in \mathbb{R}^K$
- $D = \text{diag}(\lambda_1, \cdots, \lambda_K)$. $\lambda_1, \cdots, \lambda_K \in \mathbb{R}$

PCA Step 1: PCs via Eigen-Decomposition

- Given $N \times K$ matrix $X = \begin{bmatrix} X_1^T \\ \cdots \\ X_N^T \end{bmatrix}$, reduce dimension from K to M
- Mean center the data
  1. Compute $K \times K$ covariance matrix, $S = \frac{1}{N} X^T X$
  2. Eigen-decomposition to obtain eigenvalues and eigenvectors
     - $s = UDU^T$, where $U = [u_1, \cdots, u_k]$, $D = \text{diag}(\lambda_1, \cdots, \lambda_K)$
     - Pick the M eigenvectors (principal components) with the largest eigenvalues

PCA Step 2: Project data onto PCs

$$x^T u_1 = \|x\| \|u_1\| \cos(\theta) = \|x\| \cos(\theta)$$

- u_1 is a unit vector of a principal component.
- Remember to set the center of the data to the origin!

PCA: Putting everything together

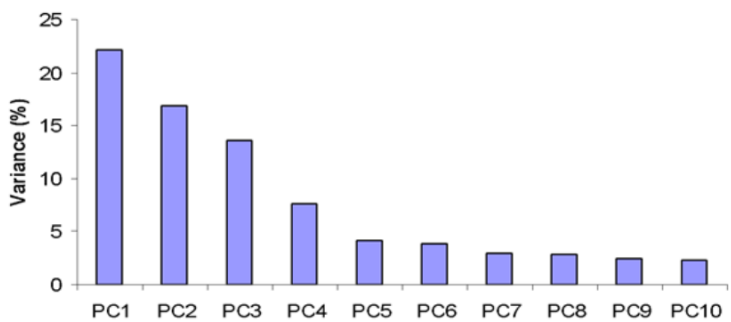
- Given $N \times K$ data matrix $K = \left[X_1^T \cdots, X_N^T\right]$, reduce dimension from K to M
- Mean center the data
- Step 1
  - Compute $K \times K$ covariance matrix $S = \frac{1}{N} X^T X$
  - Eigen-decomposition to obtain eigenvalues and eigenvectors
    - $S = UDU^T$, where $U = [u_1, \cdots, u_K]$, $D = \text{diag}(\lambda_1, \cdots, \lambda_K)$
  - Pick M eigenvectors (principal components) with the **largest eigenvalues**
- Step 2
  - Project data onto the **top** M **eigenvectors** $X_{p,i}^T = [X_i^T u_1, \cdots, X_i^T u_M]$

How Many Principal Components (PCs) M?

- The k-th eigenvalue: variance of projected data onto the k-th PC.
- The **first PC** retains the greatest fraction of **variation** in the sample
- The k-th PC retains the k-th greatest fraction of variation in the sample

Therefore,

- For K original dimensions, there are K eigenvalues and vectors. So K PCs.
- **Ignore the compoennts of lesser significance** with lower variation in data
- Strategy: Plot the eigenvalues, use only the PCs with reasonably high eigenvalues.



PCA Summary

$K \times K$ covariance matrix $S = \frac{1}{N} X^T X$

- Eigen-decomposition of $S = UDU^T$, where $U = [u_1, \cdots, u_K]$, $D = \text{diag}(\lambda_1, \cdots, \lambda_K)$, and λ_i's are sorted in descending order (λ_1 is the largest).
- $\sqrt{\lambda_k} u_k$ is the k-th PC, and u_k is called the k-th PC direction
  - Variance explained λ_k
- PC scores: projecting data onto the PC directions: $X^T_{\text{proj},i} = [X_i^T u_1, \cdots, X_i^T u_K]$
  - $\text{Var}(Xu_k) = \frac{1}{N} \sum_{i=1}^N (X_i^T u_k)^T = u_k^T (\frac{1}{N} X^T X) u_k = u_k^T S u_k = \lambda_k$
- Select top M PCs to explain a given percentage of total variance $\sum_{k=1}^K \lambda_k$

PCA as an Optimization

- Projection of vector x onto an axis u is $x^T u$
- X is a mean-centered $N \times K$ data matrix for N samples, K features
- Find a unit vector u such that the **variance of projected values is greatest**
  - Maximize $\frac{1}{N} u^T X^T X u$, such that $u^T u = 1$ (e.g, u is a unit vector).
  - Recall that

$$\text{Var}(X^T u) = \frac{1}{N} \sum_{i=1}^N (x_i^T u)^2 = \frac{1}{N} u^T X^T X u$$

How do we solve this?

- Construct Lagrangian

$$\frac{1}{N}u^T X^T X u + \lambda(1 - u^T u)$$

- Differentiate the Lagrangian with respect to u and set it to zero

$$\frac{1}{N}X^T X u - \lambda u = 0$$

- Or equivalently,

$$Su - \lambda u = 0$$

  where $S = \frac{1}{n}X^T X$: covariance matrix

- As $u \neq 0$, then u must be an eigenvector of S with eigenvalue λ

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